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|----|-----|--|-------------|----------|
| 1. | (a) | $(1.5 \times 1000 \text{ kg}) \times 4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1} \times (80 - 60) \text{ }^{\circ}\text{C} \times (1 - 15\%)$
$= 1.07 \times 10^8 \text{ J}$ | 1M+1M
1A | <u>3</u> |
| | (b) | $1.07 \times 10^8 \text{ J} \div (4.5 \text{ kW}) \div 3600 \text{ s}$
$= 6.61 \text{ (hours)}$ | 1M
1A | <u>2</u> |
| | (c) | Rate of heat transfer drops as the water temperature drops / the room temperature increases /
temperature difference drops. | 1A | <u>1</u> |

1.	(a)	5 min. (or 300 s)	1A
			1
	(b)	When the heater is switched off, its temperature is higher than the metal.	1A
		<u>Or</u> Heat continues to conduct to the metal until the same temperature is attained.	1A
		<u>Or</u> It takes time for them to attain the same temperature.	1A
			1
	(c)	(i)	1M
		$mc\Delta T = IVt$	
		$(0.80) c (45 - 20) = (4.0)(12)(5 \times 60)$	1A
		$c = 720 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$	2
		(ii)	1A
		Experimental result is higher than the actual value.	1A
		Not all the energy supplied by the heater goes to the metal.	
		<u>Or</u> Some energy is absorbed by the heater itself and / or by the thermometer.	1A
		<u>Or</u> Heat loss to the surroundings.	1A
			2
	(d)	Glass is not a good conductor of heat.	1A
		<u>Or</u> Prolonged time is required for the whole glass block to attain uniform temperature.	1A
		<u>Or</u> No insulating material is perfect, so heat loss is appreciable.	1A
			1

1. (a) - Put the sphere into the water bath for a few minutes - Put / transfer the sphere into the polystyrene cup (of water) - Measure the final / maximum temperature T_f of the water with a thermometer $0.80 \times c_b \times (80 - T_f) = 0.50 \times 4200 \times (T_f - T_0)$ $c_b = 2625 \times \frac{T_f - T_0}{80 - T_f} \text{ (J kg}^{-1} \text{ }^\circ\text{C}^{-1}\text{)}$ Precautions: - Dry the sphere with the towel quickly before putting it into the cup - Make sure the sphere is fully immersed in water - Stir the water thoroughly	1A 1A 1A 1A 1A Any TWO 1A 1A	
	6	
(b) Thermal energy / heat is lost during the transfer / drying of the sphere Or Thermal energy / heat absorbed by thermometer, stirrer or cup Or The temperature of the sphere is higher than T_f when this final temperature is measured (i.e. T_f not reaching its maximum) Thus temperature rise of water in the cup is lower than it should be.	1A 1A 1A 1A	2

Solution	Marks	Remarks
1. (a) Amount of energy required $E = mc\Delta T$ $= 6 \times 4200 \times (50 - 15)$ $= 882000 \text{ J (or 882 kJ)}$ Power $P = \frac{E}{t} = \frac{882000}{60}$ $= 14700 \text{ W (or 14.7 kW)}$	1M 1M 1A 3	
(b) Let m kg per minute be the water flow rate $mc\Delta T = Pt$ $m(4200)(40 - 15) = 14700 \times 60$ $m = 8.4 \text{ (kg min}^{-1} \text{ or kg)}$	1M 1A 2	Or $\frac{m}{t} \propto \frac{1}{\Delta T}$ $\therefore \frac{m}{6} = \frac{50 - 15}{40 - 15}$
2. (a) $n = \frac{pV}{RT} \propto pV$ (for constant T) $\frac{n_A}{n_B} = \frac{(p)(3V)}{(2p)(2V)}$ $n_A = 0.75 \times 0.8 \text{ mol}$ $= 0.6 \text{ (mol)}$	1M 1A	

1.	(a)	$5 \times 0.02 \times 3300 \times (T - 4) = 0.60 \times 4200 \times (96 - T)$ $T = 85.347368 \text{ }^{\circ}\text{C} \approx 85.3 \text{ }^{\circ}\text{C}$	1M	Accept 85.0 °C to 85.4 °C [Marking Remark (MR): temperature difference must be correct & accept missing '5' – 1 M]
			1A	
			2	
	(b)	(i)	1A	[MR: Key words - heat loss to the surroundings (from container or through evaporation)]
			1	
		(ii)	1M+1M	Assume : Power of the heater = Rate of heat loss (of the container with soup) to the surroundings [MR: $P \times t$ ✓ + 1 term ✓ - 1M $P \times t$ ✓ + 2 term ✓ - 2M $P \times t$: accept $P \times 10$ ΔT : accept (96 – 9)]
			1A	
			3	
	(iii)	Smaller than 9 °C. As the temperature of (the container with) soup drops, the rate of heat loss to the surroundings becomes lower due to a smaller temperature difference (w.r.t. the surroundings).	1A	[2 A marks are independent Key concept : smaller temperature difference]
			1A	
			2	

1.	(a)	Heat given out by iron cube = Heat gained by water $0.2 \times 450 \times (T - 33) = 0.6 \times 4200 \times (33 - 25)$ $T = 257 (^{\circ}\text{C})$	2M	
			1A	
				3
	(b)	Some water might become steam (immediately). <u>Or</u> Some water in the cup might splash.	1A	
			1A	
				1
	(c)	It is because some heat is lost to the surroundings through vaporization of water / as latent heat. <u>Or</u> It is because some heat is lost to the surroundings (or polystyrene cup) / The water might not be well stirred, otherwise, the water should have reached a temperature higher than 33°C (i.e. underestimated).	1A	
			1A	
			1A	
			1A	
				2
	(d)	(i) Poor thermal contact between the glass bob of the thermometer and the iron cube.	1A	
				1
			1A	
		(ii) Infra-red (IR) thermometer.		1

1.	(a) The glass cup absorbs thermal energy (from the hot water) / Heat is lost to the glass cup.	1A	
		1	
	(b) $4200 \times 0.4 \times (100 - 93) = H \times (93 - 23)$ $H = 168 \text{ (J } ^\circ\text{C}^{-1}) \approx 170 \text{ (J } ^\circ\text{C}^{-1})$	1M	
		1A	
		2	
	(c) (i) Infra-red (radiation) / thermal radiation	1A	
		1	
		1A	
		1	
	(ii) Remote control / infra-red (IR) thermometer / night vision camera		